



* Practical - 3 *

* Write a program to implement BFS Water Jug Problem

* water-jug.py

```
a = int(input("Enter Jug A Capacity: "))  
b = int(input("Enter Jug B Capacity: "))  
ai = int(input("Initially water in Jug A: "))  
bi = int(input("Initially water in Jug B: "))  
af = int(input("Final state of Jug A: "))  
bf = int(input("Final state of Jug B: "))
```

```
print("\nList of Operations You Can Perform:")
```

```
print("1. Fill Jug A Completely")  
print("2. Fill Jug B Completely")  
print("3. Empty Jug A Completely")  
print("4. Empty Jug B Completely")  
print("5. Pour from Jug A until Jug B is Full  
or Jug A is empty")  
print("6. Pour from Jug B until Jug A is Full  
or Jug B is empty")  
print("7. Pour all water from Jug B to Jug A")  
print("8. Pour all water from Jug A to Jug B")
```

```
while Cai != af or bi != bf :  
    op = int(input ("Enter the Operation  
                    Number: "))
```

```
    if op == 1:
```

```
        ai = a
```

```
    elif op == 2:
```

```
        bi = b
```

```
    elif op == 3:
```

```
        ai = 0
```

```
    elif op == 4:
```

```
        bi = 0
```

```
    elif op == 5:
```

```
        transfer = min (ai, b - bi)
```

```
        ai -= transfer
```

```
        bi += transfer
```

```
    elif op == 6:
```

```
        transfer = min (bi, a - ai)
```

```
        bi -= transfer
```

```
        ai += transfer
```

```
    elif op == 7:
```

```
        ai += bi
```

```
        bi = 0
```

```
    elif op == 8:
```

```
        bi += ai
```

```
        ai = 0
```

```
    else:
```

```
        print ("Invalid operation! Please enter
```

```
                a number between 1 and 8.")
```

```
print ("Current State: Jug A = {ai},
```

```
        Jug B = {bi}")
```


* Output *

Enter Jug A Capacity : 4

Enter Jug B Capacity : 3

Initially Water in Jug A : 0

Initially Water in Jug B : 0

Final State of Jug A : 2

Final State of Jug B : 0

List of Operations You can Perform:

1. Fill Jug A Completely

2. Fill Jug B Completely

3. Empty Jug A Completely

4. Empty Jug B Completely

5. Pour from Jug A until Jug B is full or Jug A is empty

6. Pour from Jug B until Jug A is full or Jug B is empty

7. Pour all water from Jug B to Jug A

8. Pour all water from Jug A to Jug B

Enter the Operation Number : 2

Current State : Jug A = 0, Jug B = 3

Enter the Operation Number : 6

Current State : Jug A = 3, Jug B = 0

Enter the Operation Number : 3

Current State : Jug A = 0, Jug B = 0